RATAN TATA MAHARASHTRA STATE SKILLS UNIVERSITY

School of Science Engineering and Technology Skills

Department of Engineering

Subject – DevOps and Automation Lab

Lab Assignment 7

Aim:

Install Jenkins and configure initial settings

Theory:

Jenkins is a widely used, open-source automation server designed to automate the building, testing, and deployment of software projects. It acts as a continuous integration and continuous delivery (CI/CD) hub, allowing developers to automate repetitive technical tasks, integrate changes frequently, and improve overall software quality.

Key Aspects of Jenkins:

- **Continuous Integration (CI):** Developers push code changes to a repository, and Jenkins immediately builds and tests the code to detect errors early.
- **Continuous Delivery (CD):** Once tested, Jenkins can automate the deployment of the software to staging or production environments.
- **Open Source and Extensible:** As an open-source project managed under the Linux Foundation, it has a vast ecosystem of plugins that allow it to integrate with nearly any tool in the DevOps toolchain (e.g., Git, Docker, Kubernetes).
- **Versatile Automation:** Beyond CI/CD, it can be used for scheduling and monitoring various tasks.

Steps:

Step 1: Install Java (Required)

Jenkins requires Java (JDK 17 or 21 recommended).

1. Download JDK 17 or 21 (Temurin / Oracle).
2. Install it.
3. Verify installation: java –version.

```
>> ~  03:22 java -version
java version "21.0.5" 2024-10-15 LTS
Java(TM) SE Runtime Environment (build 21.0.5+9-LTS-239)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.5+9-LTS-239, mixed mode, sharing)
```

Step 2: Download Jenkins

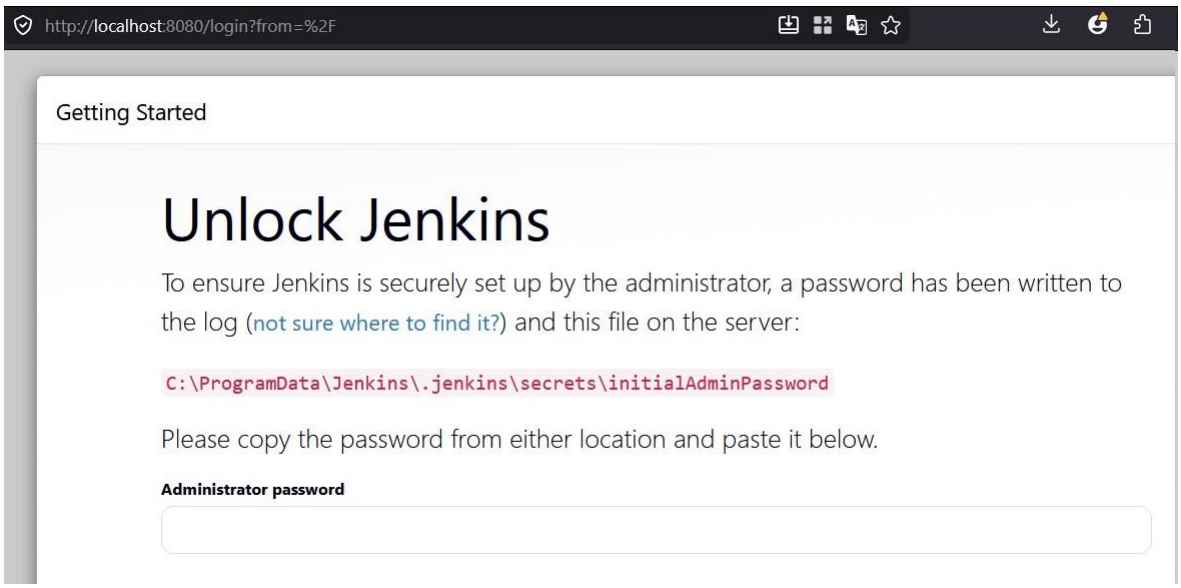
1. Go to: https://www.jenkins.io/download/
2. Choose **Windows**.
3. Download **Windows installer (.msi)**

Refer this website to install Jenkins https://www.jenkins.io/doc/book/installing/windows/.

Step 3: Access Jenkins Open browser: http://localhost:8080

You will see:

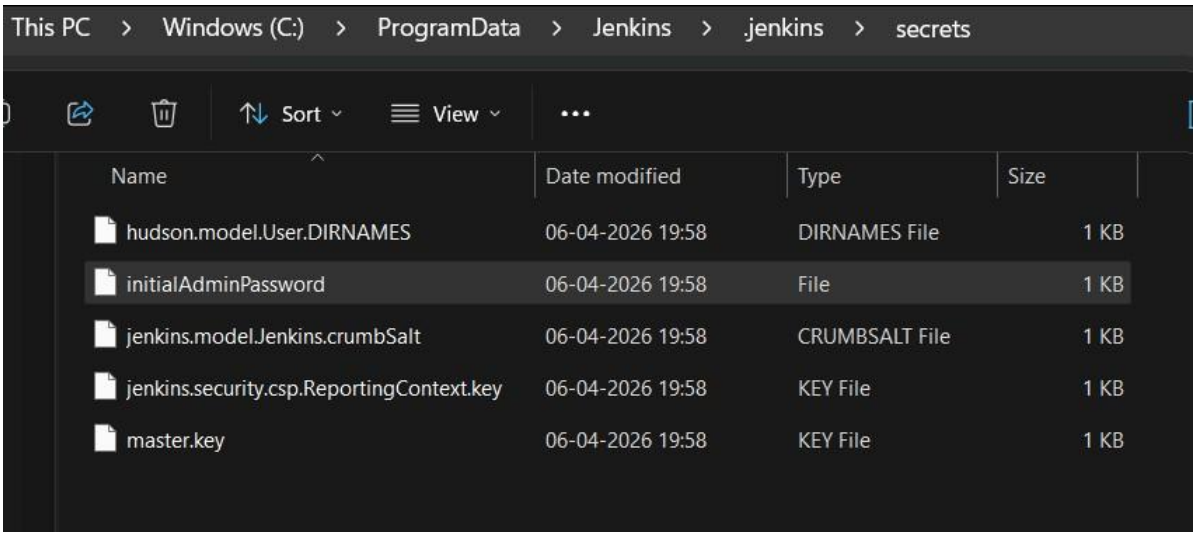
Unlock Jenkins page.



Step 4: Get Initial Admin Password

Open this file: C:\ProgramData\Jenkins\.jenkins\secrets\initialAdminPassword

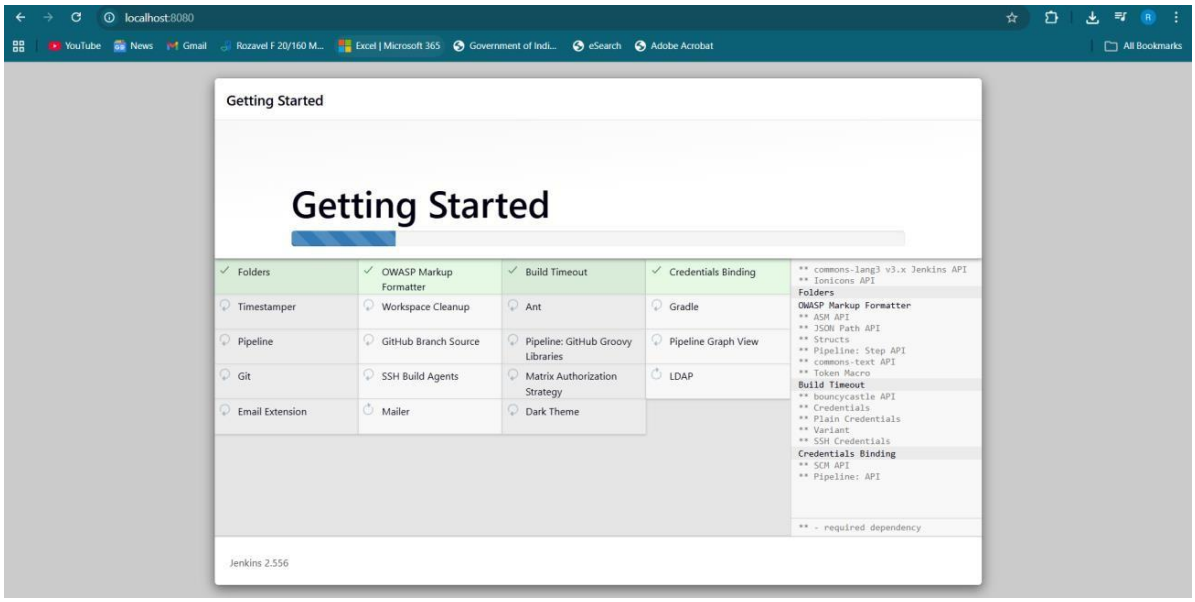
Copy the password. Paste it into browser. Click Continue.



Step 5: Install Plugins

You'll see two options: Install Suggested Plugins (Recommended) and Select plugins to install Choose: *Install Suggested Plugins*

Wait for installation to complete.



Step 6: Create Admin User

Fill:

- Username
- Password
- Full name
- Email

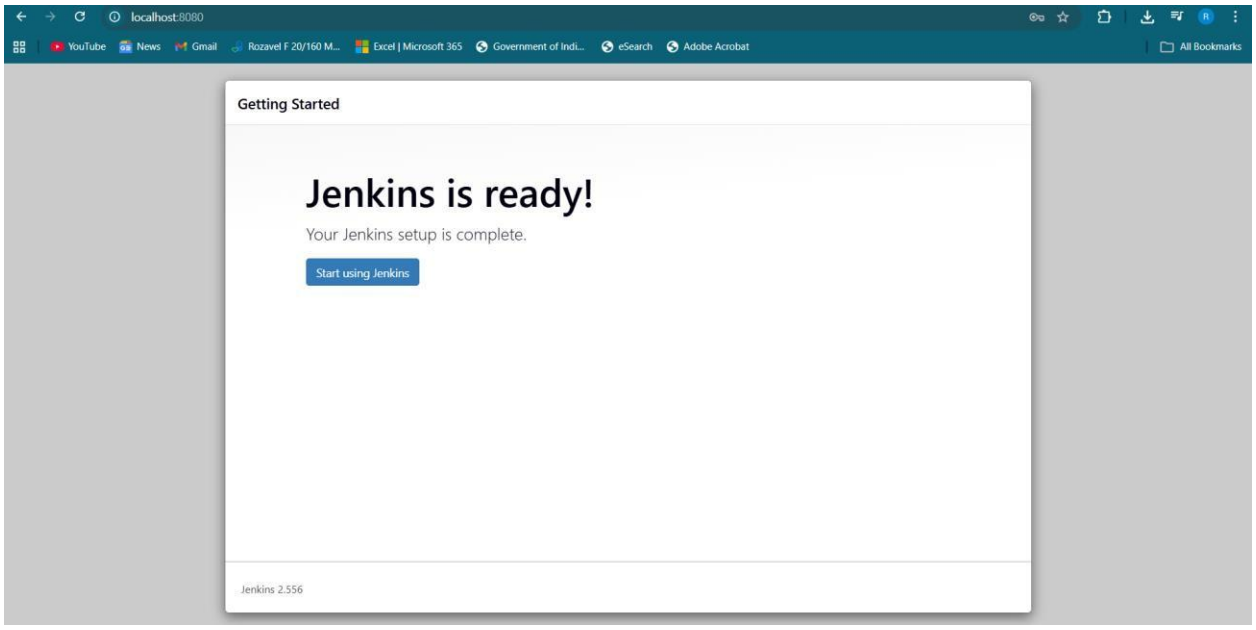
Click **Save and Continue**

Step 7: Configure Instance URL

Default: http://localhost:8080/

Click: **Save and Finish**

Click: **Start using Jenkins.**

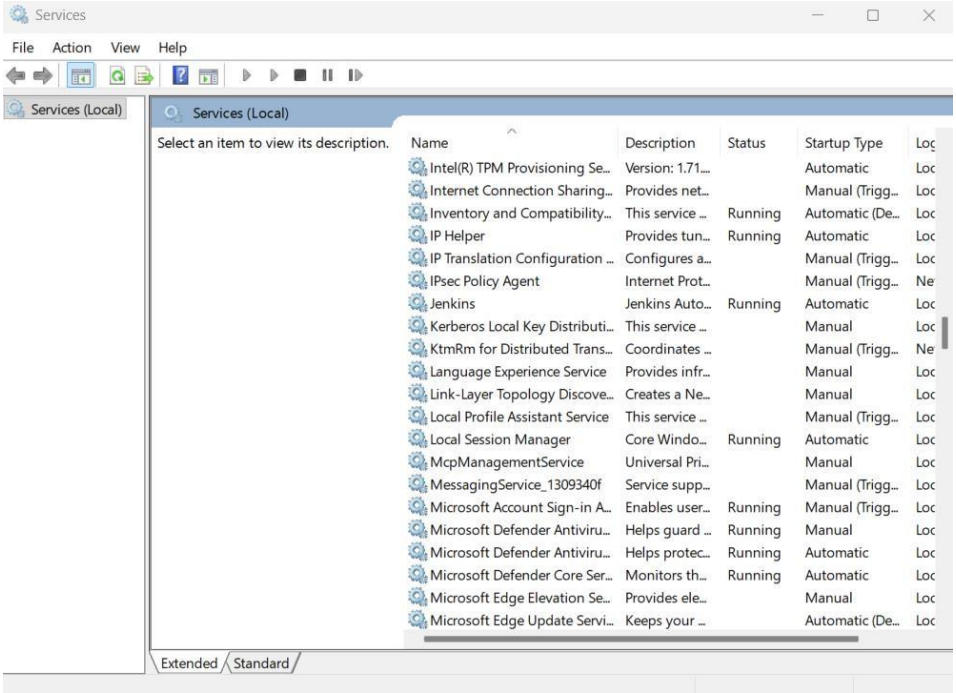


Step 8: Verify Jenkins Service

Open Command Prompt: *services.msc*

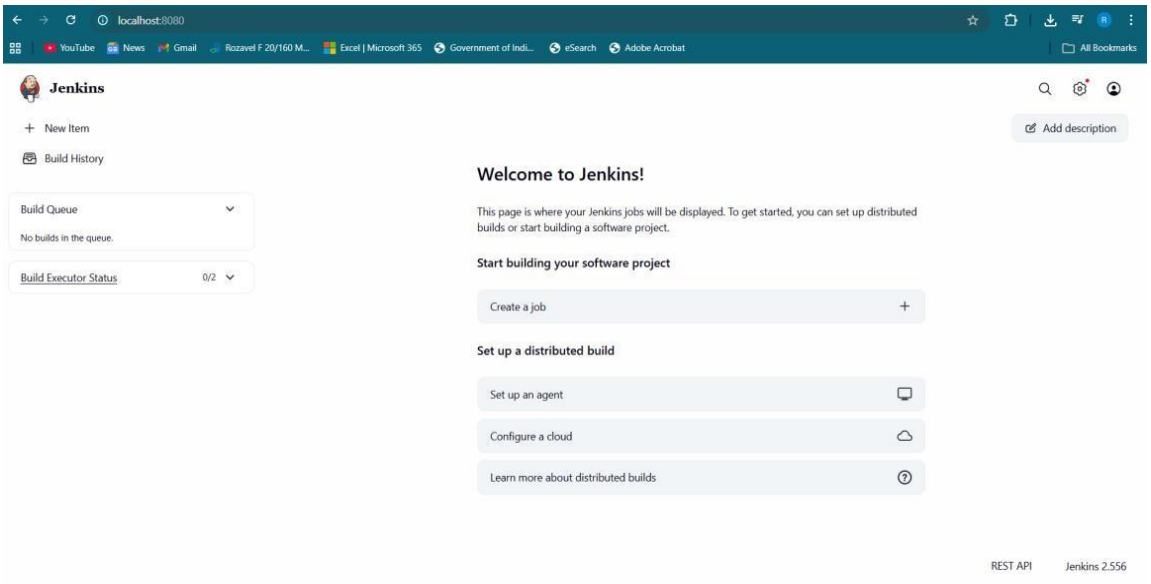
Find: *Jenkins*

Status should be: *Running*

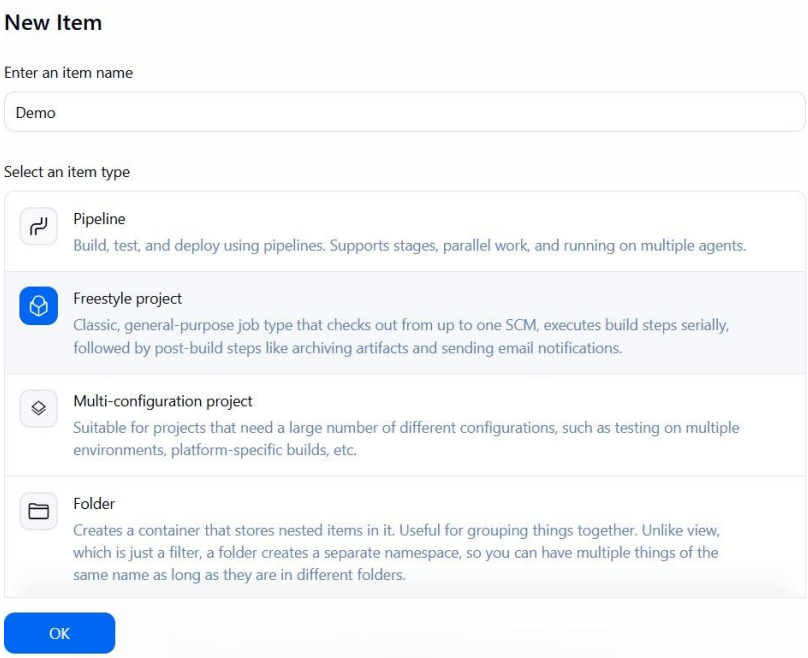


Step 9: Test Jenkins

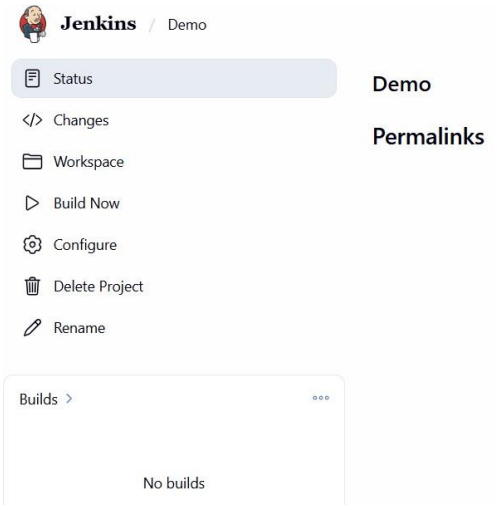
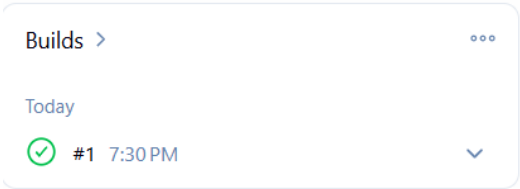
1. Click ***New Item***



2. Enter name → Choose ***Freestyle Project***
3. Click: *OK*



4. Under *Build → Add "Execute Windows Batch Command"*
5. Add: *echo Hello Jenkins*
6. Click ***Build Now***



If it runs. Jenkins is working.